MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS)
FIRST SEMESTER EXAMINATION
(2024 ADMISION ONWARDS)
MG1DSCMAT100-GROUND ROOTS OF MATHEMATICS

Duration: 2 hrs Maximum Marks: 70

*Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.*

**Part A**
**Answer any five questions. Each question carries 2 marks**

1. Define conditional statement $p \to q$ of two propositions $p$ and $q$. [K][1]
2. Give an example of a propositional function having two variables. [U][1]
3. Given $f(x) = \sqrt{x}$ and $g(x) = \sqrt{1-x}$, find $\frac{f}{g}$ and write the domain. [U][2]
4. Define odd functions. Give an example. [A][2]
5. Find $\frac{dy}{dx}$, if $y = 5x^2 + 4x + 5$. [A][4]
6. Differentiate $f(x) = x - \frac{1}{x}$. [E][4]
7. Check whether the following limits are of indeterminate form or not.
   (a) $\lim_{x\to 0^+} x \ln x$. [An][6]
   (b) $\lim_{x\to+\infty} \sqrt{x}(1-x^2)$. [An][6]
8. Find $\lim_{x\to+\infty} \frac{x}{e^x}$. [A][6]

**Part B**
**Answer any five questions. Each question carries 6 marks**

9. Explain universal quantifier, existential quantifier and the uniqueness quantifier with examples. [K][1]
10. Show that $\neg p \to (q \to r)$ and $q \to (p \vee r)$ are logically equivalent. [An][1]
11. Find the domain and graph the function $f(x) = 5 - 2x$. [U][2]

12. Verify if $f(x) = x^3 + 1$ and $g(x) = \sqrt[3]{x-1}$ are inverses of each other. [An][3]

13. Find $d^2y/dx^2$ for $y = \sin x \cos x$. [A][4]

14. Find the derivative of $\cos\left(\frac{x}{x+1}\right)$. [A][4]

15. Find the intervals in which $f(x) = x^2 - 3x + 8$ is increasing and decreasing? [A][5]

16. Find the limit $\lim_{x\to 1} \frac{x^3-x^2+x-1}{x^3-x^2}$ using L'Hôpital's rule and check the result by factoring. [A][6]

## Part C
## Answer any three questions. Each question carries 10 marks

17. (a) Show that $\forall x(P(x) \wedge Q(x))$ and $\forall x P(x) \wedge \forall x Q(x)$ are logically equivalent. [A][1]

    (b) Show that $\forall x P(x) \vee \forall x Q(x)$ and $\forall x(P(x) \vee Q(x))$ are not logically equivalent. [A][1]

18. (a) Define a one-to-one function and state the horizontal line test for one-to-one functions. [An][3]

    (b) Sketch the following functions and determine whether the following functions are one-to-one or not:
    $(i)$ $y = \sqrt{x}$ $(ii)$ $y = x^2$ [An][3]

19. (a) Find the composites $f \circ g$, $g \circ f$, $f \circ f$, and $g \circ g$ and the domain of each of the functions if $f(x) = \sqrt{x}$ and $g(x) = x + 1$. [S][2]

    (b) Given the function $y = x^2$. Write the formula when the graph of the function is shifted up 1 unit and sketch it. [S][2]

20. Use implicit differentiation to find $\frac{dy}{dx}$ if $2(x^2 + y^2)^2 = 25(x^2 - y^2)$ at $(3, 1)$. [A][4]

21. Determine the intervals where the following functions are increasing and decreasing.

    (a) $f(x) = 2x^3 - 24x + 7$. [A][5]

    (b) $f(x) = x^4 - 4x$. [A][5]

22. Identify the type of indeterminate form and evaluate the following limits.

    (a) $\lim_{x\to+\infty} x \sin\frac{\pi}{x}$ [A][6]

    (b) $\lim_{x\to\frac{\pi}{2}-} sec3x\ cos5x$. [A][6]